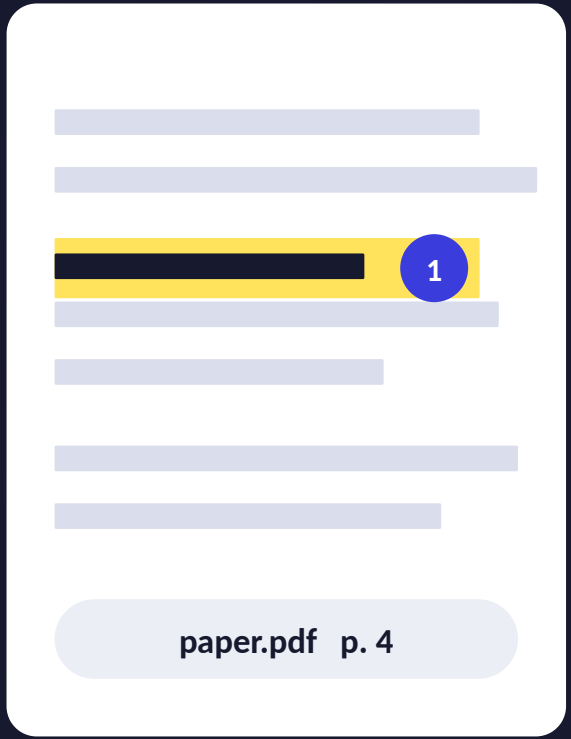
AI & Data Science | SMIT

Marginalia

Research Paper Assistant using RAG

Ask your papers. Every answer points to the page.



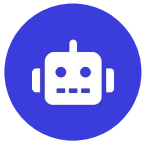
Anushay Ayat

Why papers need an assistant



Papers are long and dense

Finding the method, dataset or limitation means scanning many pages across several papers.



Plain chatbots guess

A general LLM answers from memory, so it can invent details that the paper never states.



Answers must be checkable

A useful answer shows exactly which document and page supports each claim.

What is RAG?

Retrieve

Find the passages of the uploaded papers that are most similar to the question.

Augment

Put those passages into the prompt as numbered excerpts.

Generate

The LLM writes an answer using only the excerpts and cites them.

The model reads your papers at question time instead of relying on its memory.

What the assignment asks for

- Upload one or many research PDFs
- Extract text, chunk, embed, store in a vector DB
- Chat interface for questions
- Retrieve context before answering
- Show source document and page number
- Handle empty, missing and invalid input
- Answer only from the uploaded papers
- Python, FastAPI, LangChain, ChromaDB, Gemini

Technology stack



FastAPI

Python REST API: upload, list, delete, chat.



LangChain

Text splitting, prompts and the Chroma integration.



ChromaDB

Persistent vector store with cosine similarity.



Gemini

Embeddings and answer generation.



HTML, CSS, JS

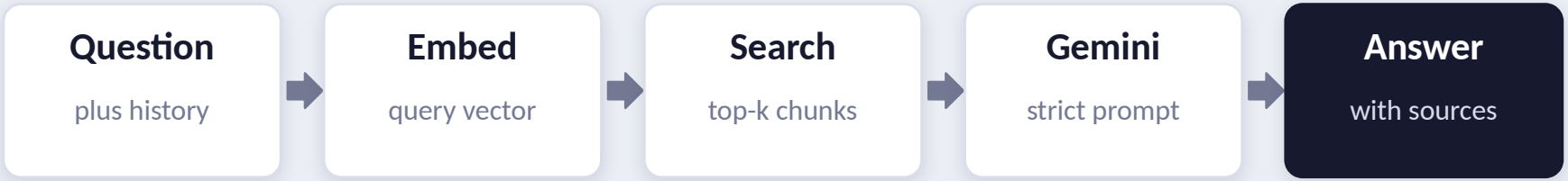
Custom frontend served by FastAPI.

System architecture

When a paper is uploaded



When a question is asked



Turning a PDF into searchable chunks

1. **Validate**
   Extension, %PDF header, size limit, empty file.

2. **Extract**
   pypdf reads every page and cleans the text.

3. **Chunk**
   1000 characters with 150 overlap, split per page.

4. **Embed and store**
   Batches of 32 with retry, saved in ChromaDB.

Stored with every chunk

```
source: "paper.pdf"
page: 4
chunk_index: 17
doc_id: "a1b2c3d4e5f6"
```

Page numbers come from this metadata, never from the model.

Retrieval and grounded answers

Retrieve first

Top-8 passages, filtered to the papers you ticked. Each paper gets its share when several are selected.

Summaries get coverage

Summary questions sample passages across the whole paper, from introduction to conclusion.

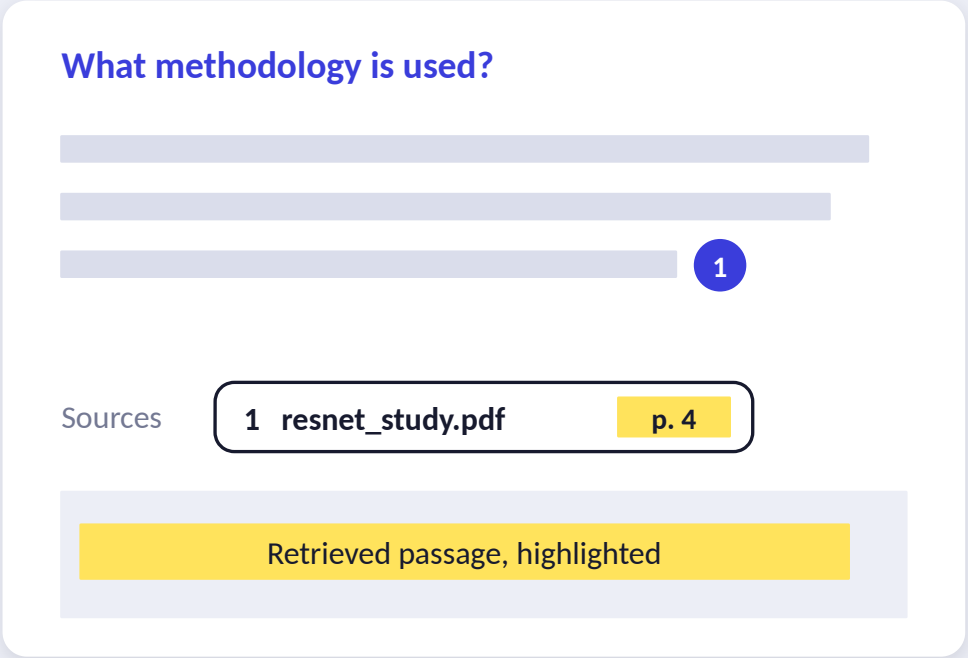
Strict prompt

Use only the excerpts, cite with [n], never add outside knowledge.

Honest not-found

If the excerpts do not cover it, the answer is: not available in the uploaded documents.

Every answer cites its source



Model cites numbers

It writes [1], [2] next to each claim.

Backend maps the numbers

Each [n] becomes file name, page and passage from stored metadata.

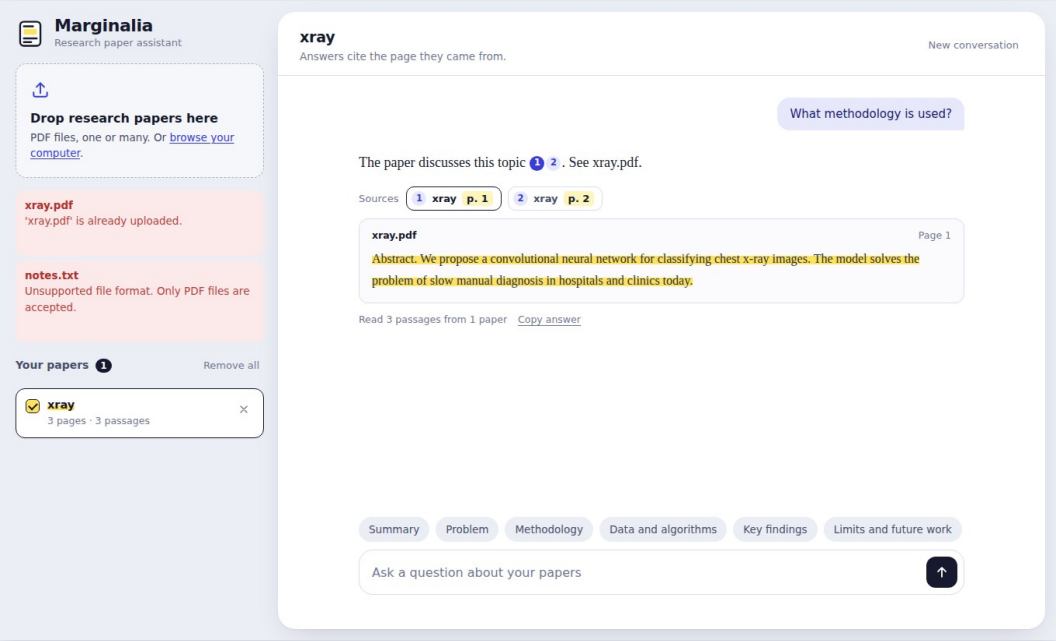
Click to verify

Opening a source shows the exact retrieved text.

Handled gracefully

Situation	What the user sees
No document uploaded	Friendly message; asking is blocked until a PDF is added
Empty question	Type a question first
Unsupported or fake PDF	That file is rejected with the reason; others still upload
Scanned, corrupt or locked PDF	Clear error; nothing is stored
Answer not in the papers	Not available in the uploaded documents
Missing API key or rate limit	Actionable message instead of a crash

The interface



Interface preview with demo data

- Drag and drop several PDFs
- Tick which papers to ask about
- One-tap starter questions
- Click a citation to see the passage
- Works on phones too

Testing

14

automated tests, all passing

Offline: fake embeddings and a fake LLM

- Upload validation: wrong type, fake PDF, corrupt PDF
- Correct file name and page in every source
- Summary requests read the whole paper
- Not-found answers return no sources
- Duplicates, deletion and multi-paper selection
- Follow-up questions and missing API key

Limitations and future work

Limitations

- Scanned PDFs need OCR, which is not included
- Tables and equations can lose their layout
- Free-tier rate limits slow very large papers

Future work

- OCR for scanned papers
- Hybrid search with re-ranking
- Streaming answers
- A retrieval-quality evaluation set
- User accounts and saved chats

Grounded, citable answers

- Answers only from your uploaded papers
- Document name and page on every answer
- Clear handling of every failure case

Thank you. Questions?